TEACHING SAM

Ask Sam to try it

SAM

Hi. They told me you are going to teach me recursion. I have genuinely never seen it before, so start wherever you like.

YOUR EXPLANATION

Explain it the way you would to someone sitting next to you

Sam only knows what you tell him.

Tell Sam

A recursive function is one that calls itself

It stops when n reaches 0 — that is the base case.

Each call works on n - 1, so the problem gets smaller.

Each call multiplies n by whatever the next call returns.

Every call keeps its own n on the stack while it waits.

WHAT SAM UNDERSTANDS

A function can call itself	Not taught
Something stops the recursion	Not taught
Each call works on a smaller problem	Not taught
Results combine as the calls return	Not taught
Each call keeps its own values	Not taught

SAM WOULD SAY IT LIKE THIS

I do not know anything about recursion yet.

TEACHING SAM

Ask Sam to try it

SAM

Hi. They told me you are going to teach me recursion. I have genuinely never seen it before, so start wherever you like.

YOUR EXPLANATION

A recursive function calls itself, and each time it works on a smaller number.

Sam only knows what you tell him.

Tell Sam

A recursive function is one that calls itself

It stops when n reaches 0 — that is the base case.

Each call works on n - 1, so the problem gets smaller.

Each call multiplies n by whatever the next call returns.

Every call keeps its own n on the stack while it waits.

WHAT SAM UNDERSTANDS

- A function can call itself — Not taught
- Something stops the recursion — Not taught
- Each call works on a smaller problem — Not taught
- Results combine as the calls return — Not taught
- Each call keeps its own values — Not taught

SAM WOULD SAY IT LIKE THIS

I do not know anything about recursion yet.

TEACHING SAM

Ask Sam to try it

YOU

A recursive function calls itself, and each time it works on a smaller number.

YOU

SAM

Hold on. If it keeps calling itself, when does it ever stop?

YOUR EXPLANATION

Explain it the way you would to someone sitting next to you

Sam only knows what you tell him.

Tell Sam

A recursive function is one that calls itself

It stops when n reaches 0 — that is the base case.

Each call works on n - 1, so the problem gets smaller.

Each call multiplies n by whatever the next call returns.

Every call keeps its own n on the stack while it waits.

WHAT SAM UNDERSTANDS

A function can call itself	Half there
Something stops the recursion	Not taught
Each call works on a smaller problem	Half there
Results combine as the calls return	Not taught
Each call keeps its own values	Not taught

SAM WOULD SAY IT LIKE THIS

A recursive function is a function that calls itself, each time on a smaller piece of the problem, *and I do not know what makes it stop.*

SAM TRIES IT: FACTORIAL(3)

```
factorial(3)  n = 3
```

WHAT SAM UNDERSTANDS

A function can call itself	Half there
Something stops the recursion	Not taught
Each call works on a smaller problem	Half there
Results combine as the calls return	Not taught
Each call keeps its own values	Not taught

SAM WOULD SAY IT LIKE THIS

A recursive function is a function that calls itself, each time on a smaller piece of the problem, *and I do not know what makes it stop.*

SAM TRIES IT: FACTORIAL(3)

```
factorial(3)  n = 3
    factorial(2)  n = 2
```

WHAT SAM UNDERSTANDS

A function can call itself	Half there
Something stops the recursion	Not taught
Each call works on a smaller problem	Half there
Results combine as the calls return	Not taught
Each call keeps its own values	Not taught

SAM WOULD SAY IT LIKE THIS

A recursive function is a function that calls itself, each time on a smaller piece of the problem, *and I do not know what makes it stop.*

SAM TRIES IT: FACTORIAL(3)

```
factorial(3)  n = 3
  factorial(2)  n = 2
    factorial(1)  n = 1
```

WHAT SAM UNDERSTANDS

A function can call itself	Half there
Something stops the recursion	Not taught
Each call works on a smaller problem	Half there
Results combine as the calls return	Not taught
Each call keeps its own values	Not taught

SAM WOULD SAY IT LIKE THIS

A recursive function is a function that calls itself, each time on a smaller piece of the problem, *and I do not know what makes it stop.*

SAM TRIES IT: FACTORIAL(3)

```
factorial(3)  n = 3
  factorial(2)  n = 2
    factorial(1)  n = 1
      factorial(0)  n = 0
```

WHAT SAM UNDERSTANDS

A function can call itself	Half there
Something stops the recursion	Not taught
Each call works on a smaller problem	Half there
Results combine as the calls return	Not taught
Each call keeps its own values	Not taught

SAM WOULD SAY IT LIKE THIS

A recursive function is a function that calls itself, each time on a smaller piece of the problem, *and I do not know what makes it stop.*

SAM TRIES IT: FACTORIAL(3)

```
factorial(3)  n = 3
  factorial(2)  n = 2
    factorial(1)  n = 1
      factorial(0)  n = 0
        factorial(-1)  n = -1
          factorial(-2)  n = -2
            factorial(-3)  n = -3
```

WHAT SAM UNDERSTANDS

A function can call itself	Half there
Something stops the recursion	Not taught
Each call works on a smaller problem	Half there
Results combine as the calls return	Not taught
Each call keeps its own values	Not taught

SAM WOULD SAY IT LIKE THIS

A recursive function is a function that calls itself, each time on a smaller piece of the problem, *and I do not know what makes it stop.*

SAM TRIES IT: FACTORIAL(3)

```
factorial(3)  n = 3
  factorial(2)  n = 2
    factorial(1)  n = 1
      factorial(0)  n = 0
        factorial(-1)  n = -1
          factorial(-2)  n = -2
            factorial(-3)  n = -3
```

SAM NEVER STOPS

You told me each call works on a smaller n, so I kept going. Nobody told me when to stop, so I did not. I ran out of room at n = -3. Where was I supposed to turn back?

WHAT SAM UNDERSTANDS

A function can call itself	Half there
Something stops the recursion	Not taught
Each call works on a smaller problem	Half there
Results combine as the calls return	Not taught
Each call keeps its own values	Not taught

SAM WOULD SAY IT LIKE THIS

A recursive function is a function that calls itself, each time on a smaller piece of the problem, *and I do not know what makes it stop.*